

# AX3工控核心板

## 规格书

文档修改历史

| 版本   | 描述 | 日期         |
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|      |    |            |
|      |    |            |

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# 产品概述

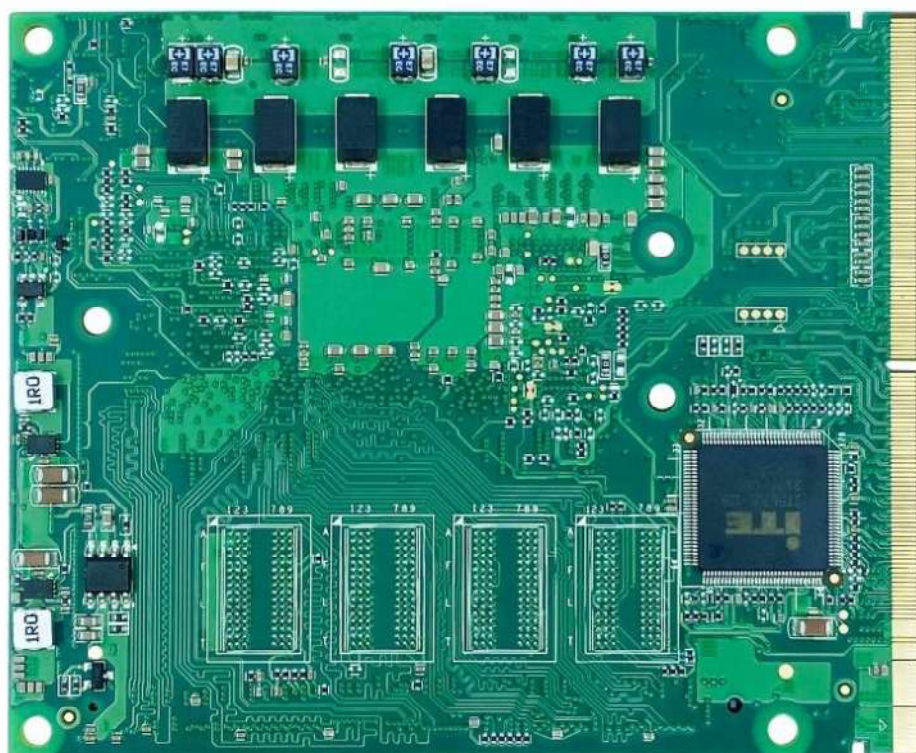
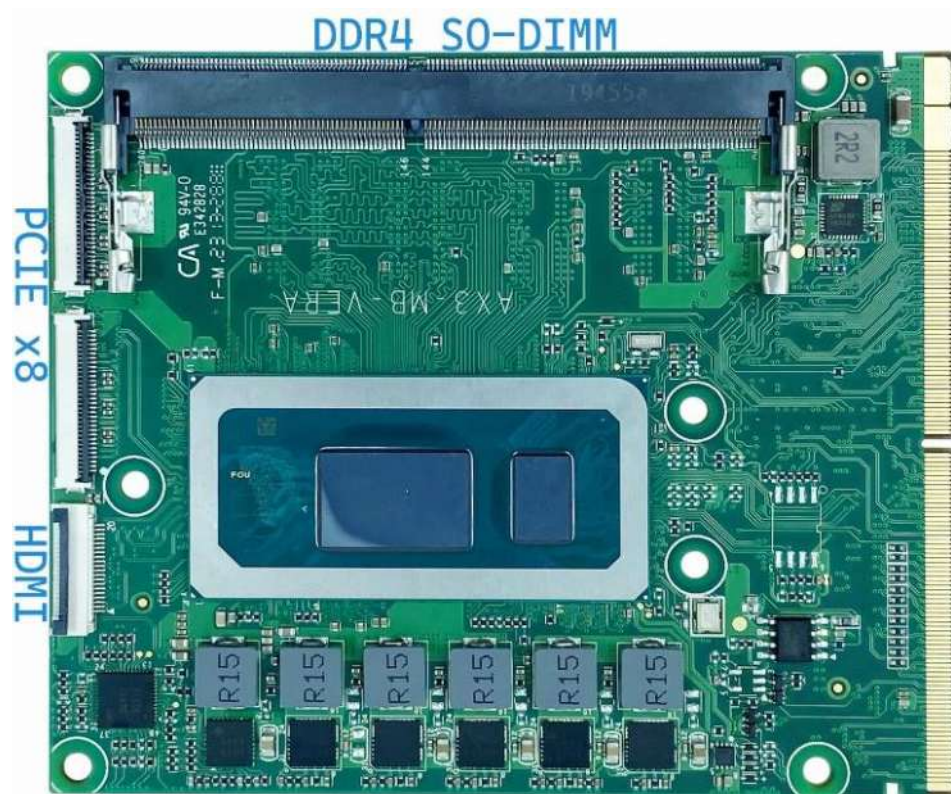
## 1.1 概述

AX3核心板采用Intel Alder Lake-P/H系列CPU，一体式MXM金手指接口设计，具有全面的功能扩展能力，满足不同用户，不同场景的使用需求。

## 1.2 特点

- ▶ 高集成度。集成 USB/LVDS/EDP/HDMI/以太网/WIFI/BT/音频等功能
- ▶ 接口丰富。USB2.0/USB3.0/SATA/PCIE/HDA/GPIO接口扩展
- ▶ 功能齐全。支持DDR板载和SODIMM选配，支持PCIE和SATA接口的SSD,支持 EDP/DP/HDMI多类型显示输出

### 1.3 外观及接口示意图

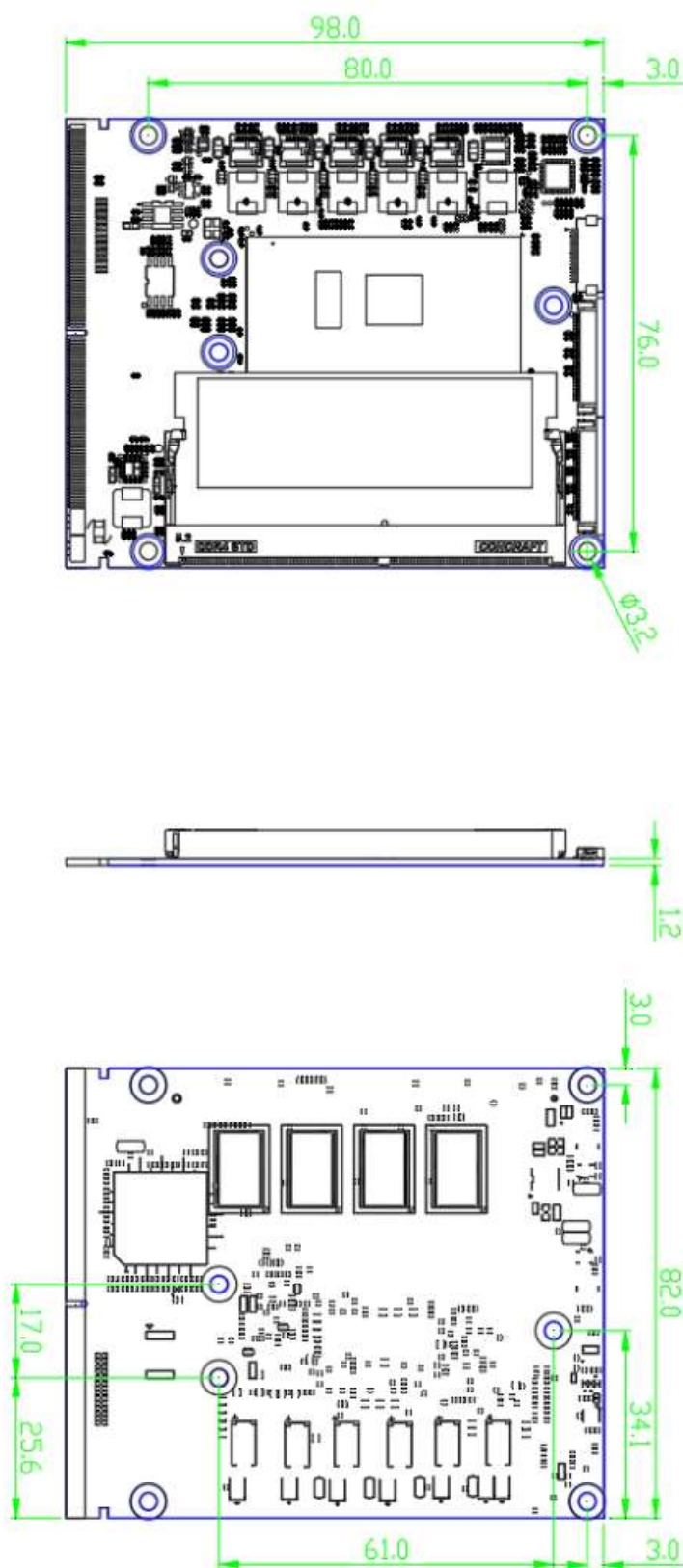


# 基本功能

|             |                 |                               |
|-------------|-----------------|-------------------------------|
| <b>CPU</b>  | <b>平台</b>       | ADL-P                         |
|             | <b>系列</b>       | 12 <sup>th</sup>              |
|             | <b>封装</b>       | BGA1744                       |
|             | <b>集成显卡</b>     | Intel® UHD/iris Graphics      |
| <b>内存</b>   | <b>类型</b>       | DDR4 MD&DDR4 SODIMM           |
|             | <b>通道</b>       | MD 和SODIMM                    |
|             | <b>容量</b>       | MD MAX 8GB,SODIMMMAX 32GB     |
| <b>外观尺寸</b> | <b>尺寸</b>       | 98*82mm                       |
| <b>信号定义</b> | <b>VIN</b>      | 9~19V 输入                      |
|             | <b>HDMI</b>     | HDMI 1.4a ,3PORT              |
|             | <b>Edp</b>      | Edp 1.4b , 4lane max4096*2304 |
|             | <b>AUDIO</b>    | HDA                           |
|             | <b>SATA</b>     | SATA3.0 2PORT                 |
|             | <b>PCIE</b>     | PCIE3.0 MAX 7PORT             |
|             | <b>USB2.0</b>   | MAX 10PORT                    |
|             | <b>USB3.0</b>   | MAX 4PORT                     |
|             | <b>ESPI</b>     | ESPI                          |
|             | <b>SPI</b>      | SPI 1PORT                     |
|             | <b>GPIO</b>     | EC GPIO ; PCH GPIO*8          |
|             | <b>I2C</b>      | 1PORT                         |
|             | <b>BEEP</b>     | PC_BEEP                       |
|             | <b>Cmos bat</b> | 3.3V BAT                      |

# 主板详细规格

## 3.1 主板尺寸图



## 3.2 接口定义

### 1、核心板mxm PIN定义

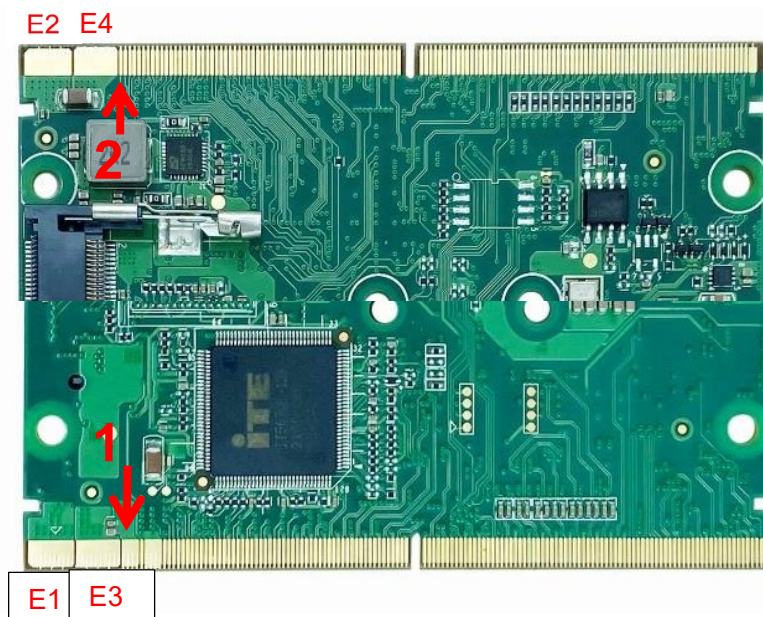
| PIN序 | 定义               | 属性        | PIN序 | 定义                 | 属性        |
|------|------------------|-----------|------|--------------------|-----------|
| E1   | VDC              | POWER/10A | E2   | VDC                | POWER/10A |
| E3   | GND              |           | E4   | GND                |           |
| 1    | 5V               | 5V/2.5A   | 2    | SUS_CLK            |           |
| 3    | 5V               |           | 4    | PEX_CLKREQ         |           |
| 5    | 5V               |           | 6    | CLK_PCIE_REQ0#     |           |
| 7    | 5V               |           | 8    | CLK_PCIE_REQ1#     |           |
| 9    | 5V               |           | 10   | SM_BAT_CLK         |           |
| 11   | GND              |           | 12   | SM_BAT_DATA        |           |
| 13   | GND              |           | 14   | AUD_BITCLK         |           |
| 15   | GND              |           | 16   | LPC_AD0            |           |
| 17   | GND              |           | 18   | PCIE_WAKE#         |           |
| 19   | BATT_IN_L        |           | 20   | DDPC_CTRLCLK       |           |
| 21   | EC_PWR_LATCH     |           | 22   | SOC_HDMI_DDC_DATA  |           |
| 23   | AUD_RESET#       |           | 24   | CLK_PCIE_REQ2#     |           |
| 25   | SOC_HDMI_DDC_CLK |           | 26   | DDPC_CTRLDATA      |           |
| 27   | eDP_HPD#         |           | 28   | EC_BKLTEN          |           |
| 29   | CHG_ACOK         |           | 30   | CLK_PCIE_WLAN_REQ# |           |
| 31   | AUD_SYNC         |           | 32   | PWRBTN#_IN#        |           |
| 33   | AUD_SDIN0        |           | 34   | PM_LANPHY_ENABLE   |           |
| 35   | AUD_SDOUT        |           | 36   | GND                |           |
| 37   | GND              |           | 38   | SPI_SCK1           |           |
| 39   | SPI_MISO1        |           | 40   | SPI_MOSI1          |           |
| 41   | FWH_WP#1         |           | 42   | V3P3A_EN           |           |
| 43   | SPI_CS1          |           | 44   | TP_DATA            |           |
| 45   | TP_CLK           |           | 46   | GND                |           |
| 47   | GND              |           | 48   | SOC_EDP_BKLTEN     |           |
| 49   | CHG_LED#         |           | 50   | CHG_FUL#           |           |
| 51   | PWR_LED#         |           | 52   | GND                |           |
| 53   | GND              |           | 54   | PCIE_LAN_WAKE#     |           |
| 55   | PCH_SML0_DAT     |           | 56   | SSD_PCIE_SATA_DET# |           |
| 57   | PCH_SML0_CLK     |           | 58   | GND                |           |
| 59   | GND              |           | 60   | TP_INT             |           |
| 61   | GPU_EVENT        |           | 62   | PCH_HDMI_HPD       |           |
| 63   | SOC_EDP_VDDEN    |           | 64   | GND                |           |
| 65   | GND              |           | 66   | DDPC_HPD           |           |
| 67   | SATA_LED#        |           | 68   | SOC_EDP_BKLTCTL    |           |
| 69   | TP_REST          |           | 70   | GND                |           |

| PIN序 | 定义                  | 属性 | PIN序 | 定义             | 属性 |
|------|---------------------|----|------|----------------|----|
| 71   | GND                 |    | 72   | BUF_PLT_RST#   |    |
| 73   | SLP_S4#             |    | 74   | ESPI_CS1       |    |
| 75   | SLP_S3#             |    | 76   | GND            |    |
| 77   | GND                 |    | 78   | DDI1_TXN3      |    |
| 79   | DDI1_TXP1           |    | 80   | DDI1_TXP3      |    |
| 81   | DDI1_TXN1           |    | 82   | GND            |    |
| 83   | GND                 |    | 84   | DDI2_TXN2      |    |
| 85   | DDI2_TXP1           |    | 86   | DDI2_TXP2      |    |
| 87   | DDI2_TXN1           |    | 88   | GND            |    |
| 89   | GND                 |    | 90   | DDI1_TXP0      |    |
| 91   | DDI2_TXP3           |    | 92   | DDI1_TXN0      |    |
| 93   | DDI2_TXN3           |    | 94   | GND            |    |
| 95   | GND                 |    | 96   | EDP_TXP1       |    |
| 97   | DDI2_TXN0           |    | 98   | EDP_TXN1       |    |
| 99   | DDI2_TXP0           |    | 100  | GND            |    |
| 101  | GND                 |    | 102  | DDI1_TXP2      |    |
| 103  | EDP_TXP2            |    | 104  | DDI1_TXN2      |    |
| 105  | EDP_TXN2            |    | 106  | GND            |    |
| 107  | GND                 |    | 108  | DDI1_AUXN      |    |
| 109  | CLK_PCIE_N0         |    | 110  | DDI1_AUXP      |    |
| 111  | CLK_PCIE_P0         |    | 112  | GND            |    |
| 113  | GND                 |    | 114  | CLK_PCIE_M2#   |    |
| 115  | EDP_TXN3            |    | 116  | CLK_PCIE_M2    |    |
| 117  | EDP_TXP3            |    | 118  | GND            |    |
| 119  | GND                 |    | 120  | CLK_PCIE_WLAN  |    |
| 121  | EDP_TXN0            |    | 122  | CLK_PCIE_WLAN# |    |
| 123  | EDP_TXP0            |    | 124  | GND            |    |
| 125  | GND                 |    | 126  | NC             |    |
| 127  | NC                  |    | 128  | NC             |    |
| 129  | NC                  |    | 130  | NC             |    |
| 131  | NC                  |    | 132  | NC             |    |
| 133  | GND                 |    | 134  | GND            |    |
| 135  | CLK_PCIE_P1         |    | 136  | DDI2_AUXP      |    |
| 137  | CLK_PCIE_N1         |    | 138  | DDI2_AUXN      |    |
| 139  | GND                 |    | 140  | GND            |    |
| 141  | CLK_PCIE_P2         |    | 142  | EDP_AUXN       |    |
| 143  | CLK_PCIE_N2         |    | 144  | EDP_AUXP       |    |
| 145  | GND                 |    | 146  | GND            |    |
| 147  | PCIE_SATA_SSD_TXP12 |    | 148  | CLK_PCIE_GPU#  |    |
| 149  | PCIE_SATA_SSD_TXN12 |    | 150  | CLK_PCIE_GPU   |    |



| PIN序 | 定义                | 属性 | PIN序 | 定义                  | 属性 |
|------|-------------------|----|------|---------------------|----|
| 151  | GND               |    | 152  | GND                 |    |
| 153  | PCIE_SSD_TXP11    |    | 154  | PCIE_SATA_SSD_RXN12 |    |
| 155  | PCIE_SSD_TXN11    |    | 156  | PCIE_SATA_SSD_RXP12 |    |
| 157  | GND               |    | 158  | PCIE_SSD_RXP11      |    |
| 159  | PCIE_SSD_TXN10    |    | 160  | PCIE_SSD_RXN11      |    |
| 161  | PCIE_SSD_TXP10    |    | 162  | PCIE_SSD_RXN9       |    |
| 163  | PCIE_SSD_TXP9     |    | 164  | PCIE_SSD_RXP9       |    |
| 165  | PCIE_SSD_TXN9     |    | 166  | GND                 |    |
| 167  | AC_POWELEV_GPIO12 |    | 168  | PCIE_SSD_RXN10      |    |
| 169  | PCIE_TXN8         |    | 170  | PCIE_SSD_RXP10      |    |
| 171  | PCIE_TXP8         |    | 172  | CLK_PCIE_M2_REQ#    |    |
| 173  | GND               |    | 174  | GND                 |    |
| 175  | SATA_TXP1         |    | 176  | PCIE_RXP8           |    |
| 177  | SATA_TXN1         |    | 178  | PCIE_RXN8           |    |
| 179  | GND               |    | 180  | GND                 |    |
| 181  | PCIE_TXP3         |    | 182  | SATA_RXP1           |    |
| 183  | PCIE_TXN3         |    | 184  | SATA_RXN1           |    |
| 185  | GND               |    | 186  | GND                 |    |
| 187  | PCIE_TXN4         |    | 188  | PCIE_RXP3           |    |
| 189  | PCIE_TXP4         |    | 190  | PCIE_RXN3           |    |
| 191  | GND               |    | 192  | GND                 |    |
| 193  | PCIE_TXN5_WLAN    |    | 194  | PCIE_RXN5_WLAN      |    |
| 195  | PCIE_TXP5_WLAN    |    | 196  | PCIE_RXP5_WLAN      |    |
| 197  | GND               |    | 198  | GND                 |    |
| 199  | PCIE_TXN6         |    | 200  | PCIE_RXP6           |    |
| 201  | PCIE_TXP6         |    | 202  | PCIE_RXN6           |    |
| 203  | GND               |    | 204  | GND                 |    |
| 205  | PCIE_TXP1         |    | 206  | PCIE_RXP4           |    |
| 207  | PCIE_TXN1         |    | 208  | PCIE_RXN4           |    |
| 209  | GND               |    | 210  | GND                 |    |
| 211  | PCIE_TXP2         |    | 212  | PCIE_RXP1           |    |
| 213  | PCIE_TXN2         |    | 214  | PCIE_RXN1           |    |
| 215  | GND               |    | 216  | GND                 |    |
| 217  | P8_USB_P8         |    | 218  | PCIE_RXP2           |    |
| 219  | P8_USB_N8         |    | 220  | PCIE_RXN2           |    |
| 221  | GND               |    | 222  | GND                 |    |
| 223  | EC_PWM5           |    | 224  | USB3_RX3_N          |    |
| 225  | HDA_SPKR          |    | 226  | USB3_RX3_P          |    |
| 227  | LPC_AD1           |    | 228  | GND                 |    |
| 229  | LPC_AD2           |    | 230  | USB3_RX2_P          |    |

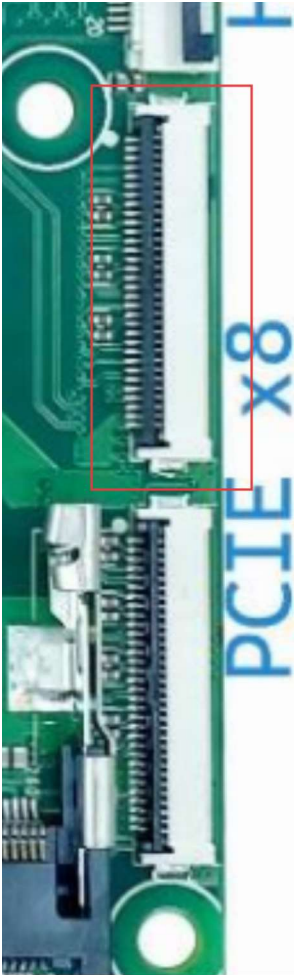
| PIN序 | 定义         | 属性 | PIN序 | 定义         | 属性      |
|------|------------|----|------|------------|---------|
| 231  | LPC_AD3    |    | 232  | USB3_RX2_N |         |
| 233  | ESPI_RESET |    | 234  | USB3_RX1_N |         |
| 235  | SYS_STAT#  |    | 236  | USB3_RX1_P |         |
| 237  | CLK_LPC_EC |    | 238  | PCICLK     |         |
| 239  | USB3_TX3_N |    | 240  | V3P3A      | 3.3V/1A |
| 241  | USB3_TX3_P |    | 242  | V3P3A      |         |
| 243  | USB3_TX2_N |    | 244  | GND        |         |
| 245  | USB3_TX2_P |    | 246  | USB3_RX4_N |         |
| 247  | USB3_TX4_N |    | 248  | USB3_RX4_P |         |
| 249  | USB3_TX4_P |    | 250  | GND        |         |
| 251  | GND        |    | 252  | P4_USB_P4  |         |
| 253  | USB3_TX1_N |    | 254  | P4_USB_N4  |         |
| 255  | USB3_TX1_P |    | 256  | GND        |         |
| 257  | GND        |    | 258  | P2_USB_P2  |         |
| 259  | P1_USB_P1  |    | 260  | P2_USB_N2  |         |
| 261  | P1_USB_N1  |    | 262  | GND        |         |
| 263  | GND        |    | 264  | P8_USB_P10 |         |
| 265  | BT_USB_P6  |    | 266  | P8_USB_N10 |         |
| 267  | BT_USB_N6  |    | 268  | GND        |         |
| 269  | GND        |    | 270  | P8_USB_N9  |         |
| 271  | CRD_USB_N7 |    | 272  | P8_USB_P9  |         |
| 273  | CRD_USB_P7 |    | 274  | P3_USB_P3  |         |
| 275  | GND        |    | 276  | P3_USB_N3  |         |
| 277  | FP_USB_P7  |    | 278  | V3P3A      | 3.3V/1A |
| 279  | FP_USB_N7  |    | 280  | V3P3A      |         |
| 281  | FAN_PWM    |    |      |            |         |



## 2、HDMI

| PIN序 | 定义        | 属性       | 描述                                                                                 |
|------|-----------|----------|------------------------------------------------------------------------------------|
| 1    | HDMI TX2P | HDMI OUT |  |
| 2    | GND       |          |                                                                                    |
| 3    | HDMI TX2N |          |                                                                                    |
| 4    | HDMI TX1P |          |                                                                                    |
| 5    | GND       |          |                                                                                    |
| 6    | HDMI TX1N |          |                                                                                    |
| 7    | HDMI TX0P |          |                                                                                    |
| 8    | GND       |          |                                                                                    |
| 9    | HDMI_TX0N |          |                                                                                    |
| 10   | HDMI_CLKP |          |                                                                                    |
| 11   | GND       |          |                                                                                    |
| 12   | HDMI_CLKN |          |                                                                                    |
| 13   | GND       |          |                                                                                    |
| 14   | GND       |          |                                                                                    |
| 15   | HDMID_SCL |          |                                                                                    |
| 16   | HDMID_SDA |          |                                                                                    |
| 17   | GND       |          |                                                                                    |
| 18   | 5V        |          |                                                                                    |
| 19   | HDMI_HPD  |          |                                                                                    |
| 20   | GND       |          |                                                                                    |

### 3、PCIE X8 接口定义 (ADL-H系列支持)

| PIN序 | 定义             | 属性      | 描述                                                                                  |
|------|----------------|---------|-------------------------------------------------------------------------------------|
| 1    | GND            | PCIE X4 |  |
| 2    | PEG_TXN_3      |         |                                                                                     |
| 3    | PEG_TXP_3      |         |                                                                                     |
| 4    | GND            |         |                                                                                     |
| 5    | PEG_RXN_3      |         |                                                                                     |
| 6    | PEG_RXP_3      |         |                                                                                     |
| 7    | GND            |         |                                                                                     |
| 8    | PEG_TXN_2      |         |                                                                                     |
| 9    | PEG_TXP_2      |         |                                                                                     |
| 10   | GND            |         |                                                                                     |
| 11   | PEG_RXN_2      |         |                                                                                     |
| 12   | PEG_RXP_2      |         |                                                                                     |
| 13   | GND            |         |                                                                                     |
| 14   | PEG_TXN_1      |         |                                                                                     |
| 15   | PEG_TXP_1      |         |                                                                                     |
| 16   | GND            |         |                                                                                     |
| 17   | PEG_RXN_1      |         |                                                                                     |
| 18   | PEG_RXP_1      |         |                                                                                     |
| 19   | GND            |         |                                                                                     |
| 20   | PEG_TXN_0      |         |                                                                                     |
| 21   | PEG_TXP_0      |         |                                                                                     |
| 22   | GND            |         |                                                                                     |
| 23   | PEG_RXN_0      |         |                                                                                     |
| 24   | PEG_RXP_0      |         |                                                                                     |
| 25   | GND            |         |                                                                                     |
| 26   | CLK_PCIE_PEG#2 |         |                                                                                     |
| 27   | CLK_PCIE_PEG2  |         |                                                                                     |
| 28   | GND            |         |                                                                                     |
| 29   | GND            |         |                                                                                     |
| 30   | BUF_PLT_RST#   |         |                                                                                     |
| 31   | GND            |         |                                                                                     |
| 32   | V3P3A          | 3.3V A  |                                                                                     |

| PIN序 | 定义        | 属性      | 描述 |
|------|-----------|---------|----|
| 1    | GND       | PCIE X4 |    |
| 2    | PEG_TXN_7 |         |    |
| 3    | PEG_TXP_7 |         |    |
| 4    | GND       |         |    |
| 5    | PEG_RXN_7 |         |    |
| 6    | PEG_RXP_7 |         |    |
| 7    | GND       |         |    |
| 8    | PEG_TXN_6 |         |    |
| 9    | PEG_TXP_6 |         |    |
| 10   | GND       |         |    |
| 11   | PEG_RXN_6 |         |    |
| 12   | PEG_RXP_6 |         |    |
| 13   | GND       |         |    |
| 14   | PEG_TXN_5 |         |    |
| 15   | PEG_TXP_5 |         |    |
| 16   | GND       |         |    |
| 17   | PEG_RXN_5 |         |    |
| 18   | PEG_RXP_5 |         |    |
| 19   | GND       |         |    |
| 20   | PEG_TXN_4 |         |    |
| 21   | PEG_TXP_4 |         |    |
| 22   | GND       |         |    |
| 23   | PEG_RXN_4 |         |    |
| 24   | PEG_RXP_4 |         |    |
| 25   | GND       |         |    |
| 26   | GND       |         |    |
| 27   | GND       |         |    |
| 28   | GND       |         |    |
| 29   | V3P3S     | 3.3V S  |    |
| 30   | V3P3S     |         |    |
| 31   | V3P3S     |         |    |
| 32   | V3P3S     |         |    |